ResilienceProject

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Time Budget Data from Two Farms

Material

Two commercial farms, one in Sweden (Farm A) and one in The Netherlands (Farm B). Data collected on 15 newly introduced heifers from each farm. The data also contains records from first lactating cows that have been in the barn for 100 days.

Functions

First I make some functions for computing 5-day means, standard deviations, and for plotting:

```
filter_summarise <- function(cow.data, hours.threshold=16) {
  den = 1
  test = cow.data$hours<=1
  if (all(test)) den=24
  cow.data$budget[which(cow.data$hours < hours.threshold/den)]=NA
  cow.data$Period = 1
  cow.data$Period[cow.data$day>20 & cow.data$day <26]= NA
  cow.data$Period[cow.data$day>25]= 2
  df = cow.data %>%
    group_by(Area, Period) %>%
    summarize(avg_time_per_day=mean(budget, na.rm=TRUE), proportion_NA = sum(is.na(budget))/le
    na.omit()
  print(df)
  return(cow.data)
}

Overall_GAMLSS <- function(cow.data, farm) {
  results = list()
  kk=0
  for (AREA in c("Cubicle", "Feed", "Alley")) {
  df1 <- cow.data[cow.data$Area==AREA,]
  df10 <- na.omit(df1)
  if (farm=="Farm A") mod.group<-gamlss(budget~cs(day)*factor(ind)*factor(Period) +
                                          random(factor(date)) +n_intro + hours + trimming,
                                        sigma.fo=~day + n_intro  + hours + trimming,
                                        data=df10, control=gamlss.control(trace=FALSE),
                                          method=mixed(20,20))
  if (farm=="Farm B") mod.group<-gamlss(budget~cs(day)*factor(ind)*factor(Period) +
                                          random(factor(date)) +n_intro + hours + factor(ind),
                                        sigma.fo=~day + n_intro  + hours,
                                        data=df10, control=gamlss.control(trace=FALSE),
                                          method=mixed(20,20))
  model_summary <- capture.output(summary(mod.group))
  print("########################################")
```

```
  print(AREA)
  print("Estimates of nuisance parameters not shown")
  test1 <- substr(model_summary, 1,7) == "n_intro"
  test2 <- test1 + 1*(substr(model_summary, 1,5) == "hours")
  test3 <- test2 + 1*(substr(model_summary, 1,8) == "trimming")
  test4 <- test3 + 1*(substr(model_summary, 1,3) == "day")
  test5 <- test4 + 1*(substr(model_summary, 1,2) == "Mu")
  test6 <- test5 + 1*(substr(model_summary, 1,5) == "Sigma")
  test7 <- test6 + 1*(substr(model_summary, 1,7) == "(Interc")
  #for (i in 7:(length(model_summary)-13)) cat(model_summary[i],"\n")
  for (i in which(test7==1)) cat(model_summary[i],"\n")
  kk= kk+1
  results[[kk]] = mod.group
  }

  return(results)
}

plot_GAMLSS_Results <- function(Results_GAMLSS, Heifers, farm) {
  kk = 0
  for (AREA in c("Cubicle", "Feed", "Alley")) {
    df1 <- Heifers[Heifers$Area==AREA,]
    df10 = na.omit(df1)
    kk = kk + 1
    mod.group = Results_GAMLSS[[kk]]
    y.all = NULL
    for (IND in unique(df1$ind)) {
      df11 = df10[df10$ind==IND,]
      df11$n_intro=0
      df11$hours=24
      if (farm=="Farm A") df11$trimming=0
      df12 =na.omit(df11)
      df12$date = df12$date[1]
      y=predict(mod.group, what="mu", newdata=df12, data=df10)
      y.all <- c(y.all, y)
    }
    df.all <- cbind(na.omit(df1), y.all)
    p <- ggplot(df.all, aes(day, y.all)) +
      geom_point() +
      geom_point(x=df.all$day, y=df.all$budget ,
                 color="red") +
      facet_wrap(~ind) +
      ggtitle(paste(farm, AREA)) +
      ylab("Proportion of day") +
      ylim(0,1)
    print(p)
  }
}

get.5dMean <- function(cow.data, area="Cubicle", remove5=FALSE) {
  m <- numeric(9)
  cow.data = cow.data[cow.data$Area==area,]
  for (i in 1:9) m[i] <- mean(cow.data[cow.data$day %in% (i*5-4):(i*5),]$budget, na.rm =TRUE)
  if (remove5) m <- m[-5]
```

```
  return(m)
}

Get.SDs <- function(cow.data, area="Cubicle", days=c(1,20)) {
  n = length(unique(cow.data$ind))
  vec.days <- days[1]:days[2]
  vec.sd <- numeric(n)
  for (id in 1:n) {
    vec.sd[id] = sd(cow.data[cow.data$Area==area & cow.data$ind==id & cow.data$day %in% vec.da
  }
  return(vec.sd)
}

Get.SDs_residuals <- function(cow.data, area="Cubicle", days=c(1,20)) {
  n = length(unique(cow.data$ind))
  vec.days <- days[1]:days[2]
  vec.sd <- numeric(n)
  for (id in 1:n) {
    y = cow.data[cow.data$Area==area & cow.data$ind==id & cow.data$day %in% vec.days,]$budget
    x = cow.data[cow.data$Area==area & cow.data$ind==id & cow.data$day %in% vec.days,]$day
    df1 = data.frame(x=x, y=y)
    mod1 = gamlss(y~cs(x), sigma.fo=~1, data=na.omit(df1), control=gamlss.control(trace=FALSE)
    res1 = mod1$mu.fv - df1$y
    vec.sd[id] = sd(res1, na.rm=TRUE)
  }
  return(vec.sd)
}

violin_plots <- function(Heifers, CowDIM100, Area_use ) {
  SD.df = data.frame(SD=c(Get.SDs(Heifers, area=Area_use), Get.SDs(Heifers,days=c(26,45), area
                    Age=c(rep("Heifers", n1*2), rep("Older cows DIM>100", n2*2)),
                    Period=c(rep("Day 1-20", n1), rep("Day 26-45",n1),rep("Day 1-20", n2), re
  )
  rubrik=paste(Area_use,"Area")
  ggplot(SD.df, aes(x=Age,y=SD)) +
    geom_violin() +
    geom_violin(draw_quantiles = c(0.5)) +
    geom_point(alpha=0.5) +
    facet_wrap(~Period)+
    xlab("") +
    ylab("Standard deviations") +
    ggtitle(rubrik)
}

plot_smooth <- function(cow.data, AREA = "Cubicle", days=1:20, smooth.method="gam", main=NULL
  df1 = cow.data[cow.data$Area==AREA,]
  df1 = df1[df1$day %in% days,]
  if (is.null(main)) main=paste("Days",days[1],"to",days[length(days)])
  ggplot(df1, aes(x=day, y=budget)) +
    geom_point() +
    facet_wrap(~ind) +
    geom_smooth(method=smooth.method) +
    ylab("Proportion of the day") +
```

```
    ggtitle(main)
}
```

Analysis and plotting of data

Farm A

First we analyze Farm A

```
farm="Farm A"
#hours.threshold = 16
CowDIM100 <- read.csv(paste0(data.path,"CowDIM100Group1 (2).txt"), sep="")
CowDIM100 <- filter_summarise(CowDIM100)
```

```
`summarise()` has grouped output by 'Area'. You can override using the
`.groups` argument.

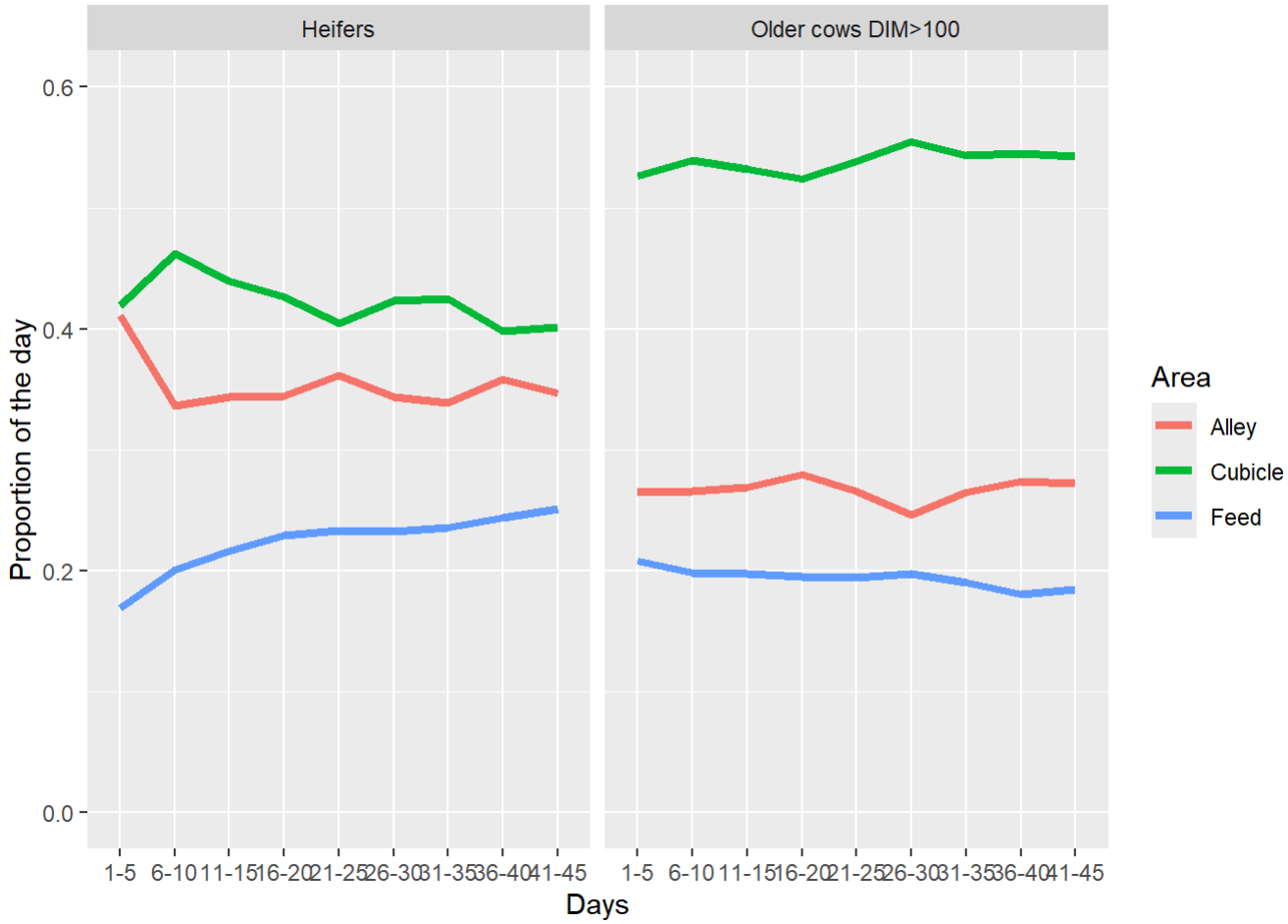
# A tibble: 6 × 4
# Groups:   Area [3]
  Area    Period avg_time_per_day proportion_NA
  <chr>    <dbl>            <dbl>         <dbl>
1 Alley        1            0.270        0.05
2 Alley        2            0.264        0.0458
3 Cubicle      1            0.530        0.05
4 Cubicle      2            0.547        0.0458
5 Feed         1            0.200        0.05
6 Feed         2            0.188        0.0458
```

```
Heifers<- read.csv(paste0(data.path,"BudgetData_w_hours_trimming.txt"), sep="")
Heifers <- filter_summarise(Heifers)
```

```
`summarise()` has grouped output by 'Area'. You can override using the
`.groups` argument.

# A tibble: 6 × 4
# Groups:   Area [3]
  Area    Period avg_time_per_day proportion_NA
  <chr>    <dbl>            <dbl>         <dbl>
1 Alley        1            0.359        0.0333
2 Alley        2            0.347        0.0433
3 Cubicle      1            0.437        0.0333
4 Cubicle      2            0.412        0.0433
5 Feed         1            0.204        0.0333
6 Feed         2            0.241        0.0433
```

Make plots to get an overall picture per group (i.e. Heifers and older cows):



Overall GAMLSS modell

```
Results_GAMLSS =Overall_GAMLSS(Heifers, farm)
```

```
Warning in summary.gamlss(mod.group): summary: vcov has failed, option qr is used instead

[1] "########################################"
[1] "Cubicle"
[1] "Estimates of nuisance parameters not shown"
Mu link function:  identity
Mu Coefficients:
(Intercept)                          0.4445575  0.0805428   5.520 5.49e-08
n_intro                             -0.0084400  0.0042849  -1.970 0.049429
hours                                0.0031241  0.0029724   1.051 0.293761
trimming                             0.0111979  0.0224167   0.500 0.617625
(Intercept)                         ***
n_intro                             *
hours
trimming
Sigma link function:  log
Sigma Coefficients:
(Intercept) -1.693465   0.571849  -2.961  0.00319 **
day         -0.004952   0.002167  -2.285  0.02267 *
n_intro      0.119238   0.042642   2.796  0.00534 **
hours       -0.042626   0.024016  -1.775  0.07644 .
trimming     0.057960   0.253099   0.229  0.81895
```

```
Warning in summary.gamlss(mod.group): summary: vcov has failed, option qr is used instead

[1] "########################################"
[1] "Feed"
[1] "Estimates of nuisance parameters not shown"
Mu link function:  identity
Mu Coefficients:
(Intercept)                 2.059e-01  2.254e-02   9.138  < 2e-16
n_intro                    -2.329e-03  1.338e-03  -1.741 0.082335
hours                       1.787e-03  8.218e-04   2.175 0.030163
trimming                   -8.964e-03  6.040e-03  -1.484 0.138449
(Intercept)                ***
n_intro                    .
hours                      *
trimming
Sigma link function:  log
Sigma Coefficients:
(Intercept) -3.944294   0.571849  -6.897 1.41e-11 ***
day          0.006920   0.002167   3.194  0.00148 **
n_intro     -0.082221   0.042642  -1.928  0.05433 .
hours        0.003883   0.024016   0.162  0.87160
trimming    -0.579276   0.253099  -2.289  0.02246 *

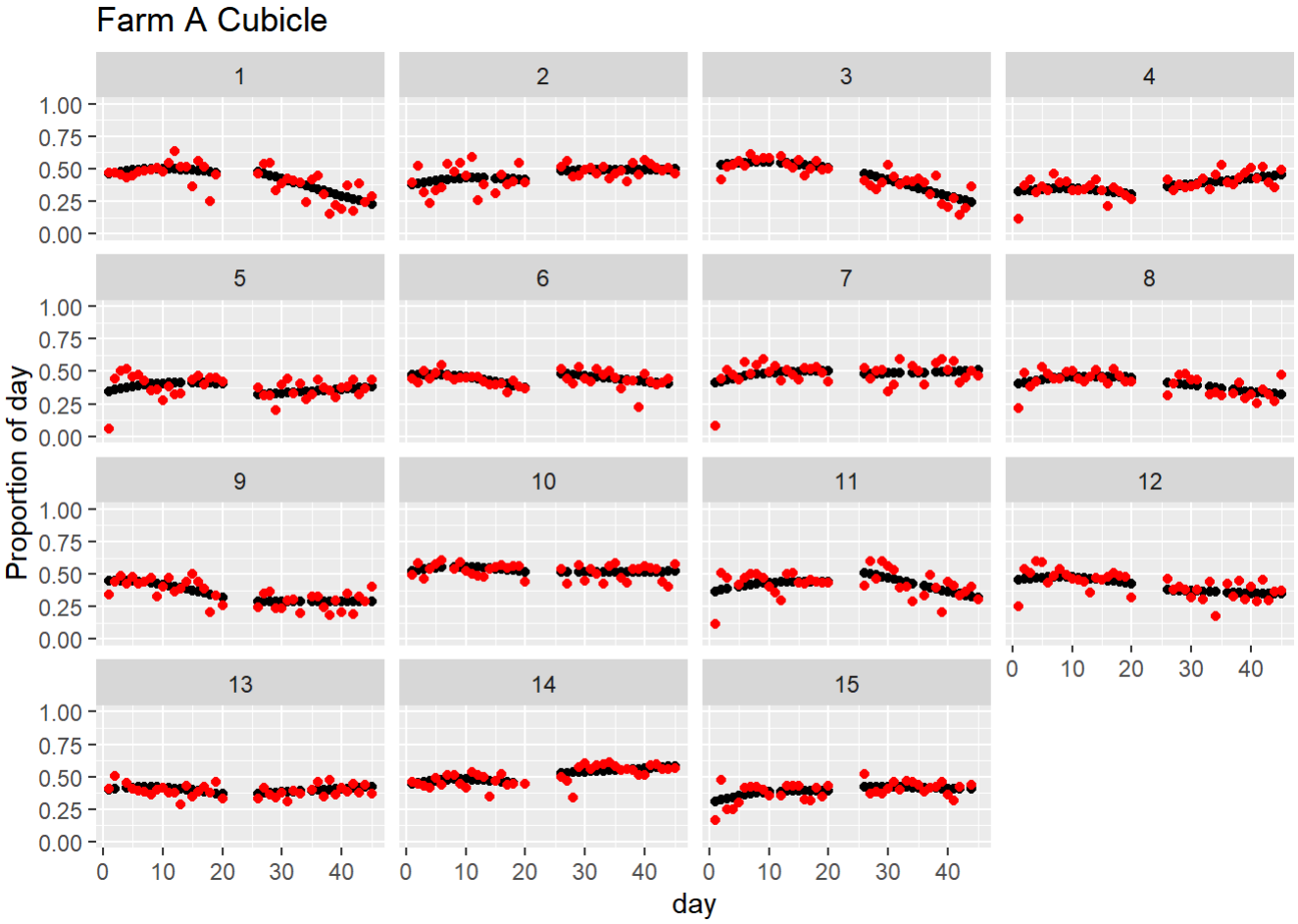
Warning in summary.gamlss(mod.group): summary: vcov has failed, option qr is used instead

[1] "########################################"
[1] "Alley"
[1] "Estimates of nuisance parameters not shown"
Mu link function:  identity
Mu Coefficients:
(Intercept)                 0.3263223  0.0711778   4.585 5.82e-06
n_intro                     0.0099810  0.0041775   2.389 0.017273
hours                      -0.0039108  0.0025325  -1.544 0.123193
trimming                   -0.0007379  0.0152375  -0.048 0.961396
(Intercept)                ***
n_intro                    *
hours
trimming
Sigma link function:  log
Sigma Coefficients:
(Intercept) -2.234472   0.571849  -3.907 0.000104 ***
day         -0.005491   0.002167  -2.534 0.011541 *
n_intro      0.100084   0.042642   2.347 0.019261 *
hours       -0.019482   0.024016  -0.811 0.417592
trimming    -0.393358   0.253099  -1.554 0.120697
```

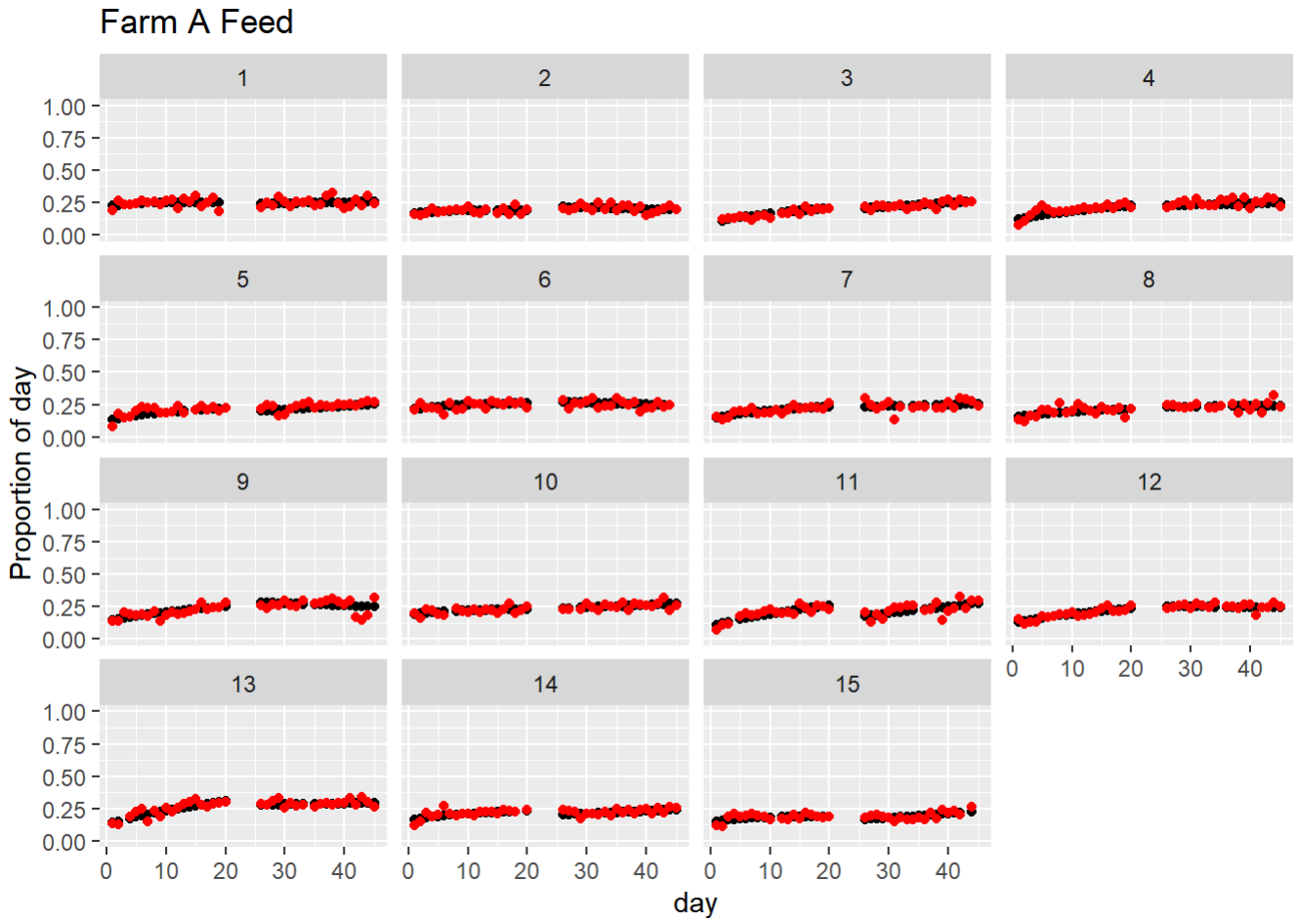
```
plot_GAMLSS_Results(Results_GAMLSS, Heifers, farm)
```

```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```

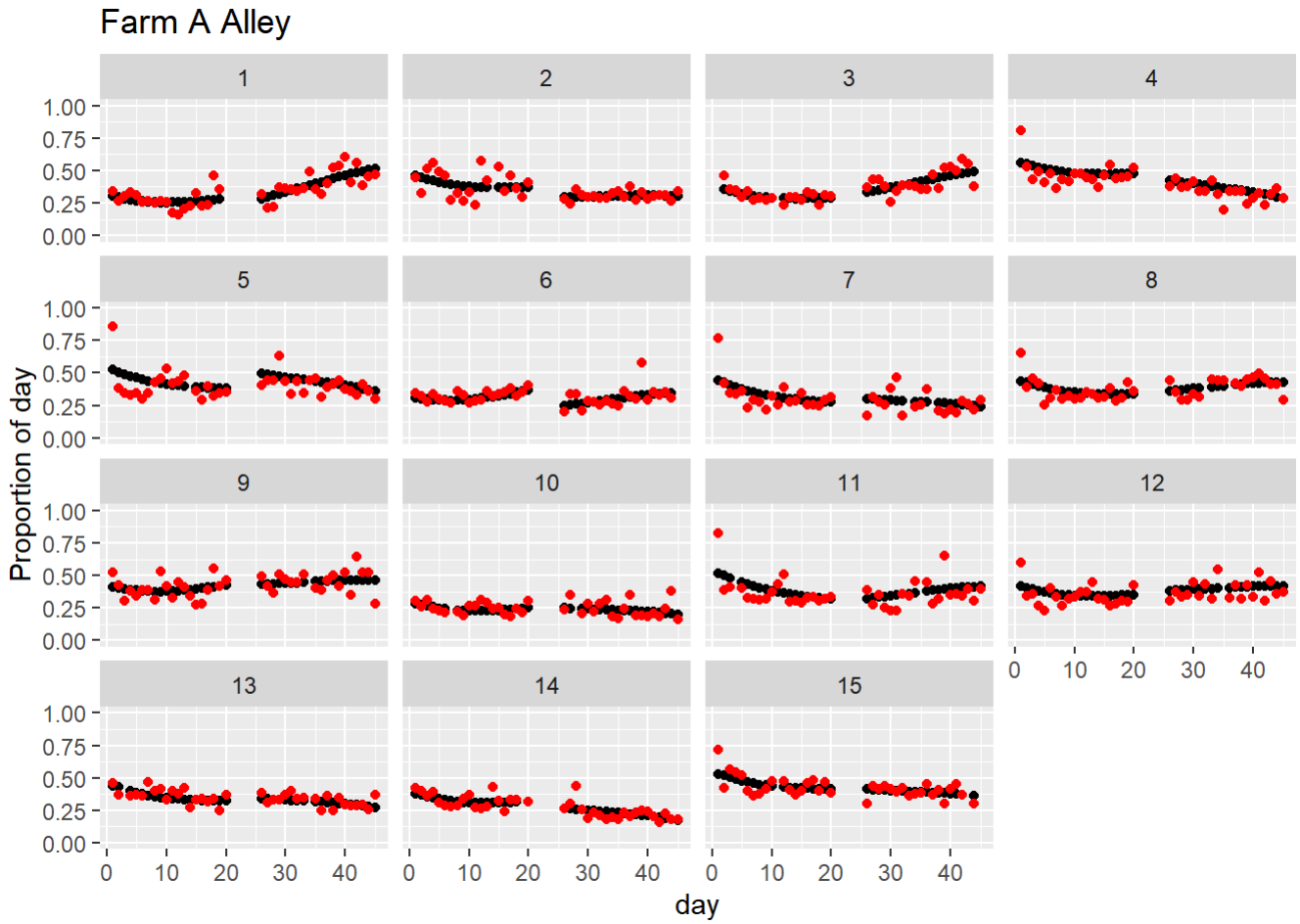
```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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```




```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```



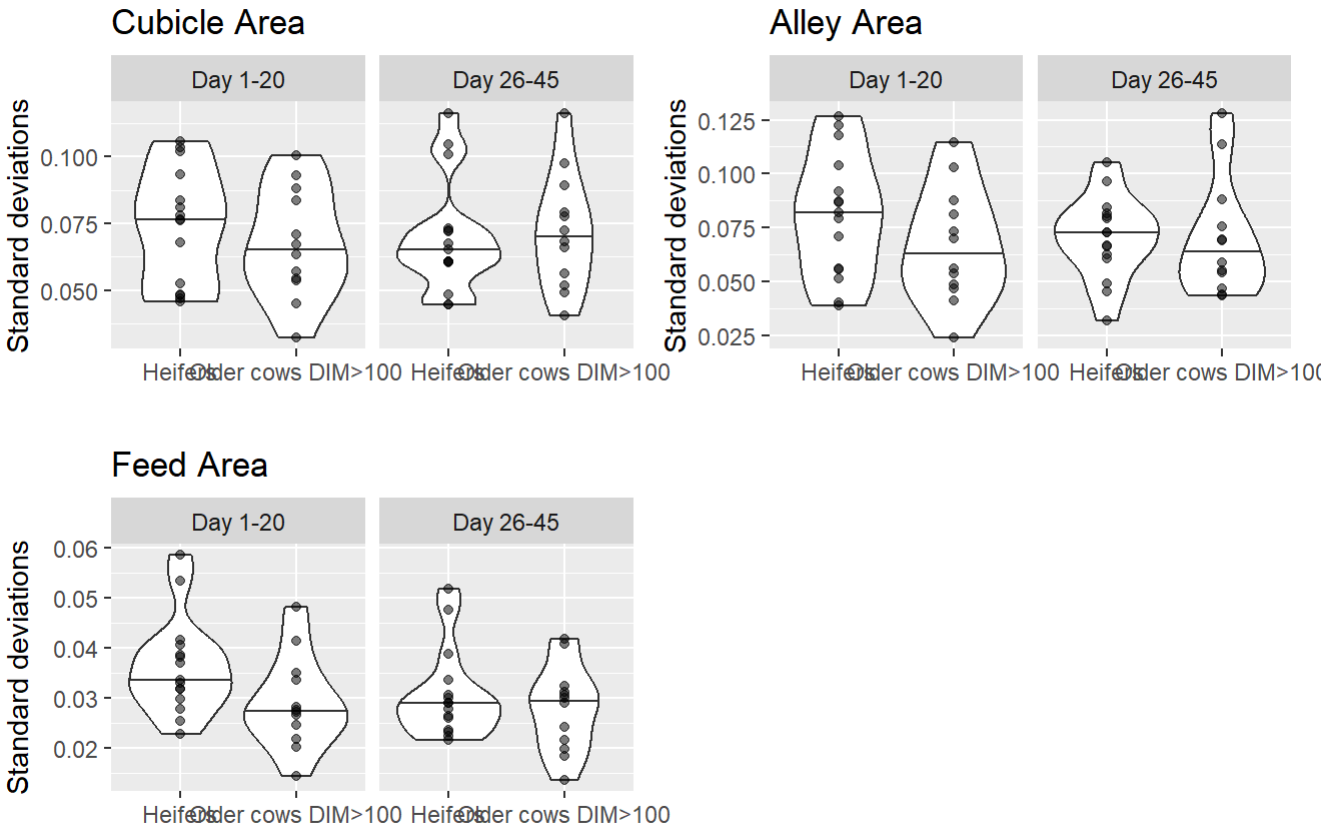

```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```



Violin plots

Here we compute the variation of time spent in an area type between days (for a specified time period) for each individual separately and then plotting these as a violin plot:

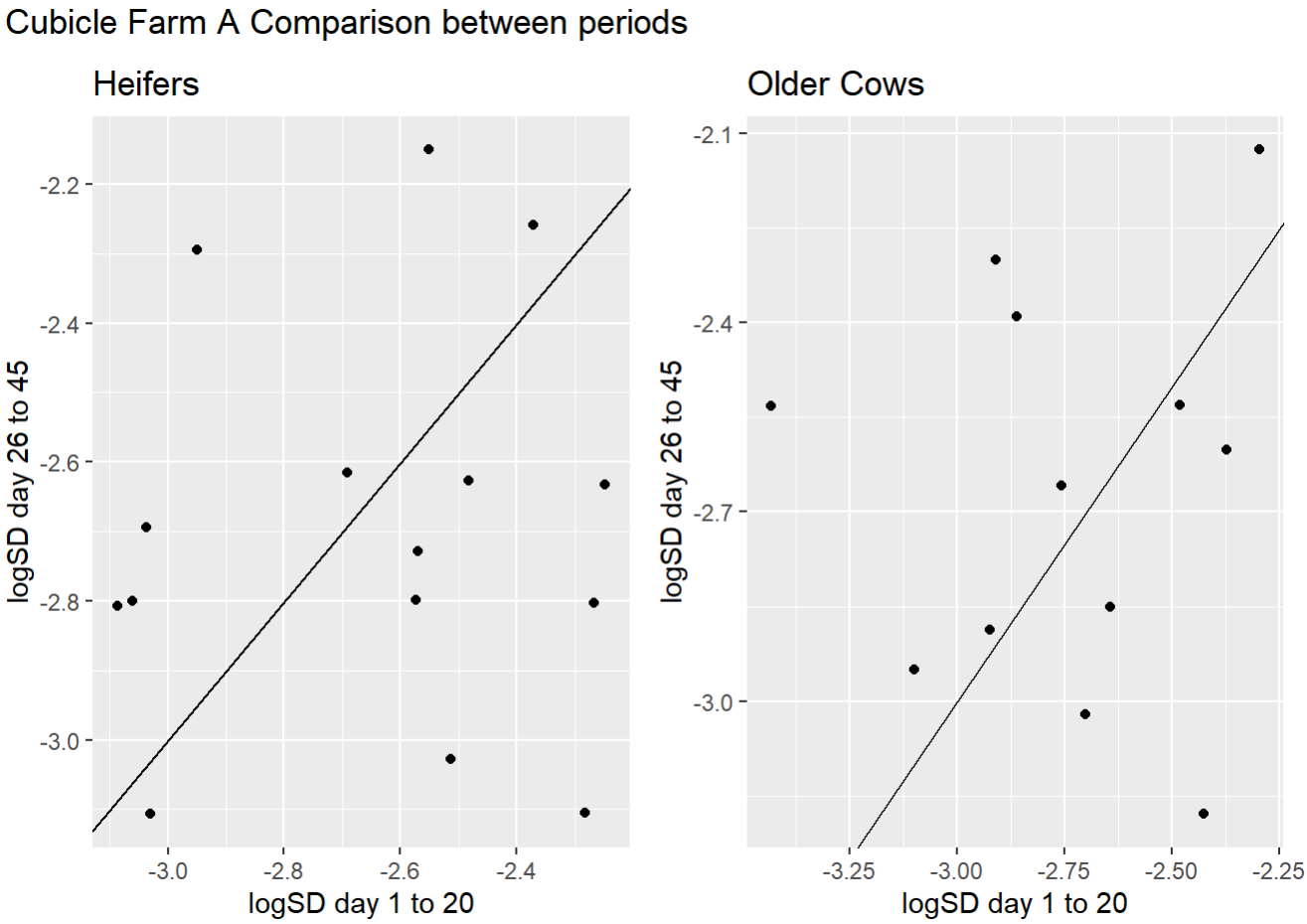
Farm A



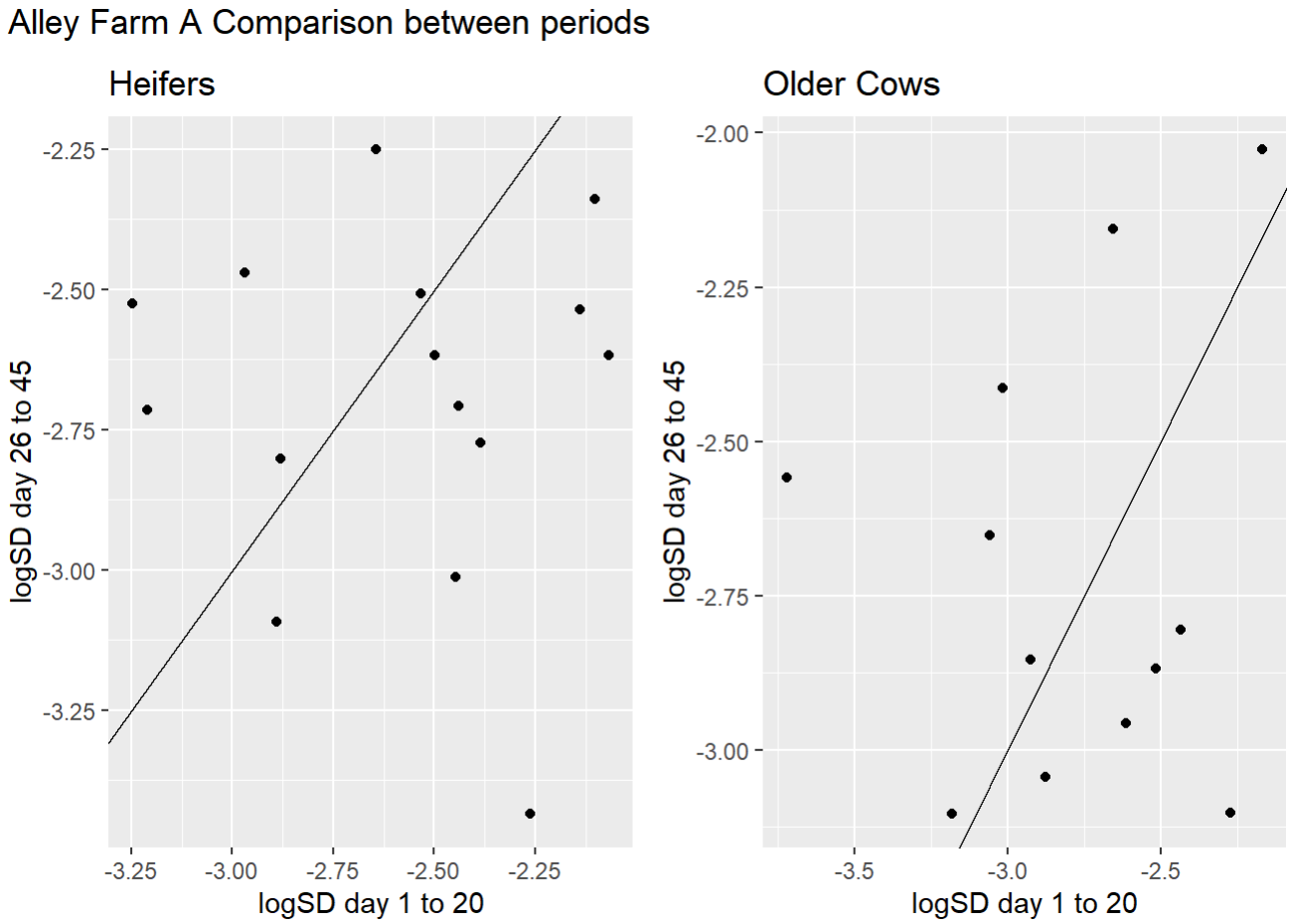
Analysis and plots of standard deviations

Here we analyze and plot the standard deviation of the raw data. We use log-transform to make it more normally distributed.

```
 Cubicle
Difference Period 1: P-value= 0.451
Difference Period 2: P-value= 0.814
Difference between Period 1 and 2 for Heifers: P-value= 0.648
```




```
 Alley
Difference Period 1: P-value= 0.211
Difference Period 2: P-value= 0.887
Difference between Period 1 and 2 for Heifers: P-value= 0.397
```

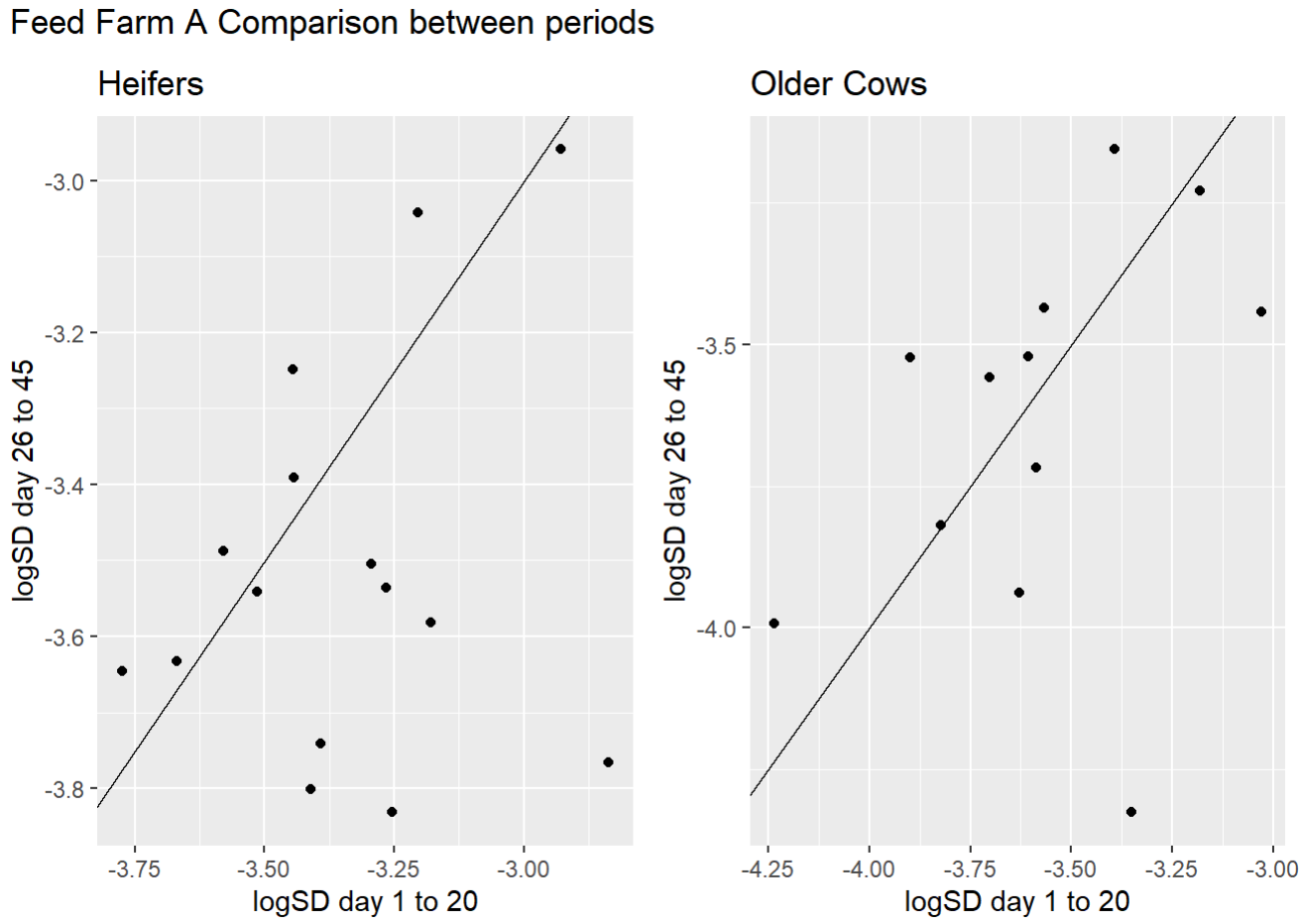


Feed

Difference Period 1: P-value= 0.05

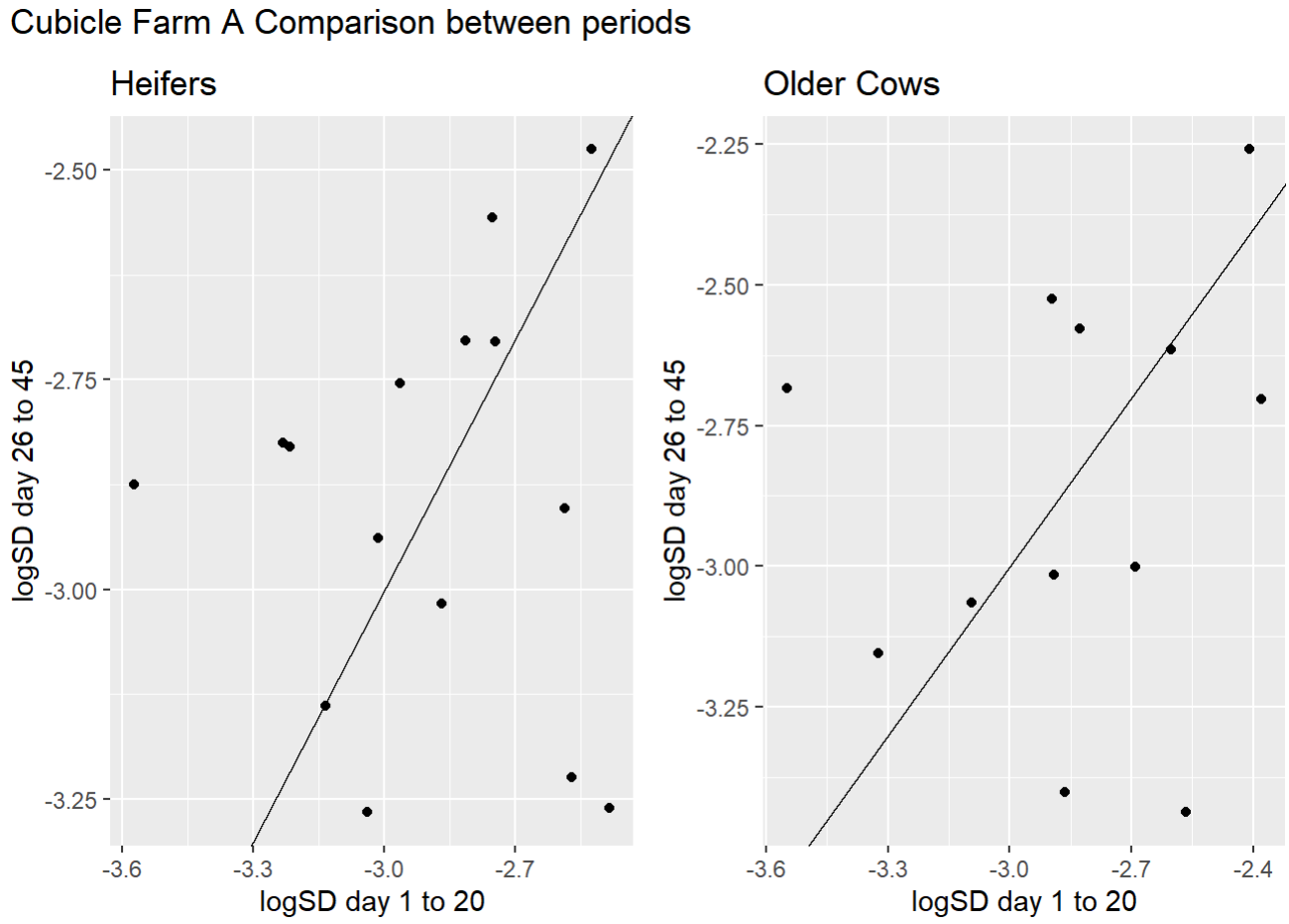
Difference Period 2: P-value= 0.305

Difference between Period 1 and 2 for Heifers: P-value= 0.061

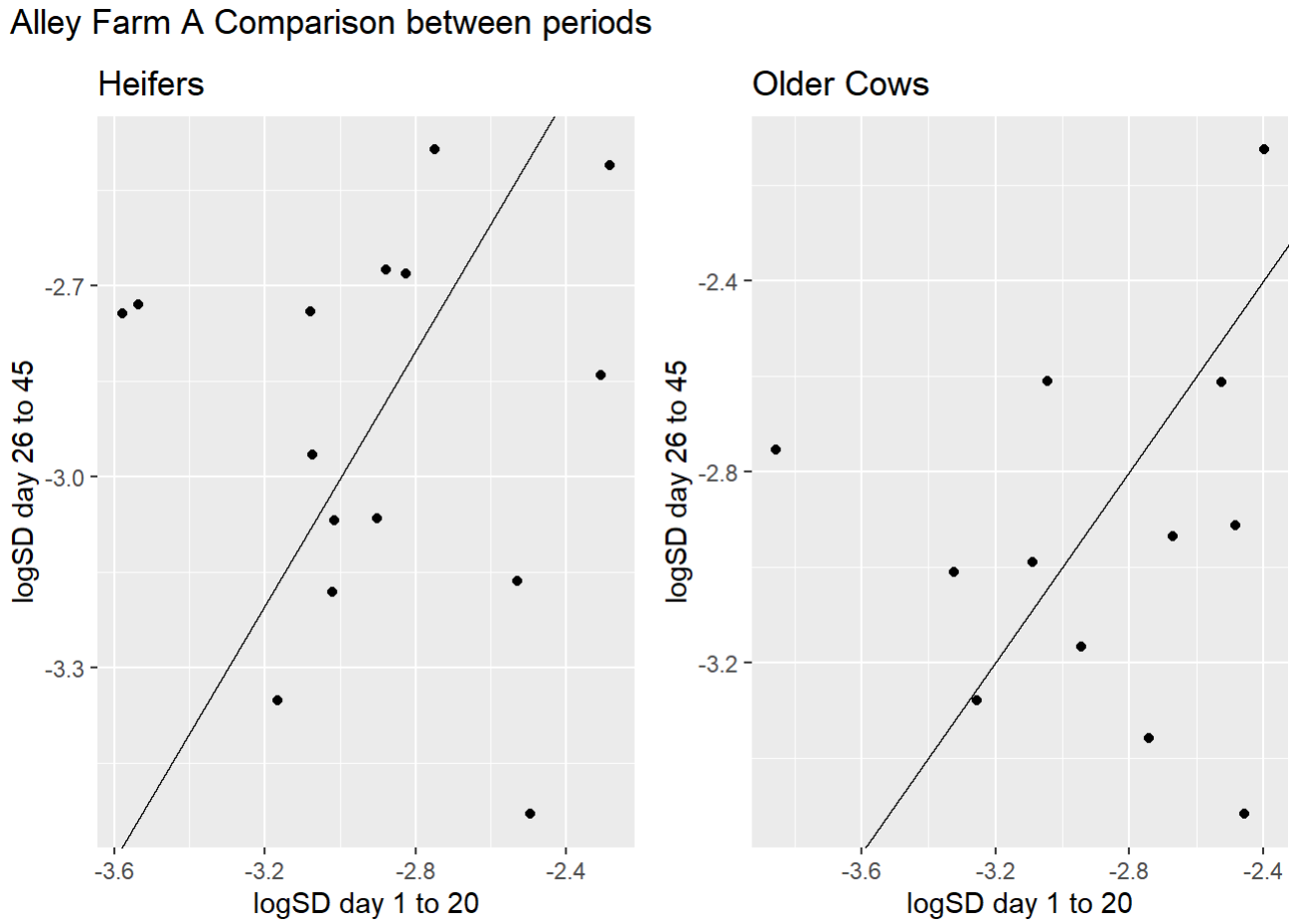


SD of the residuals:

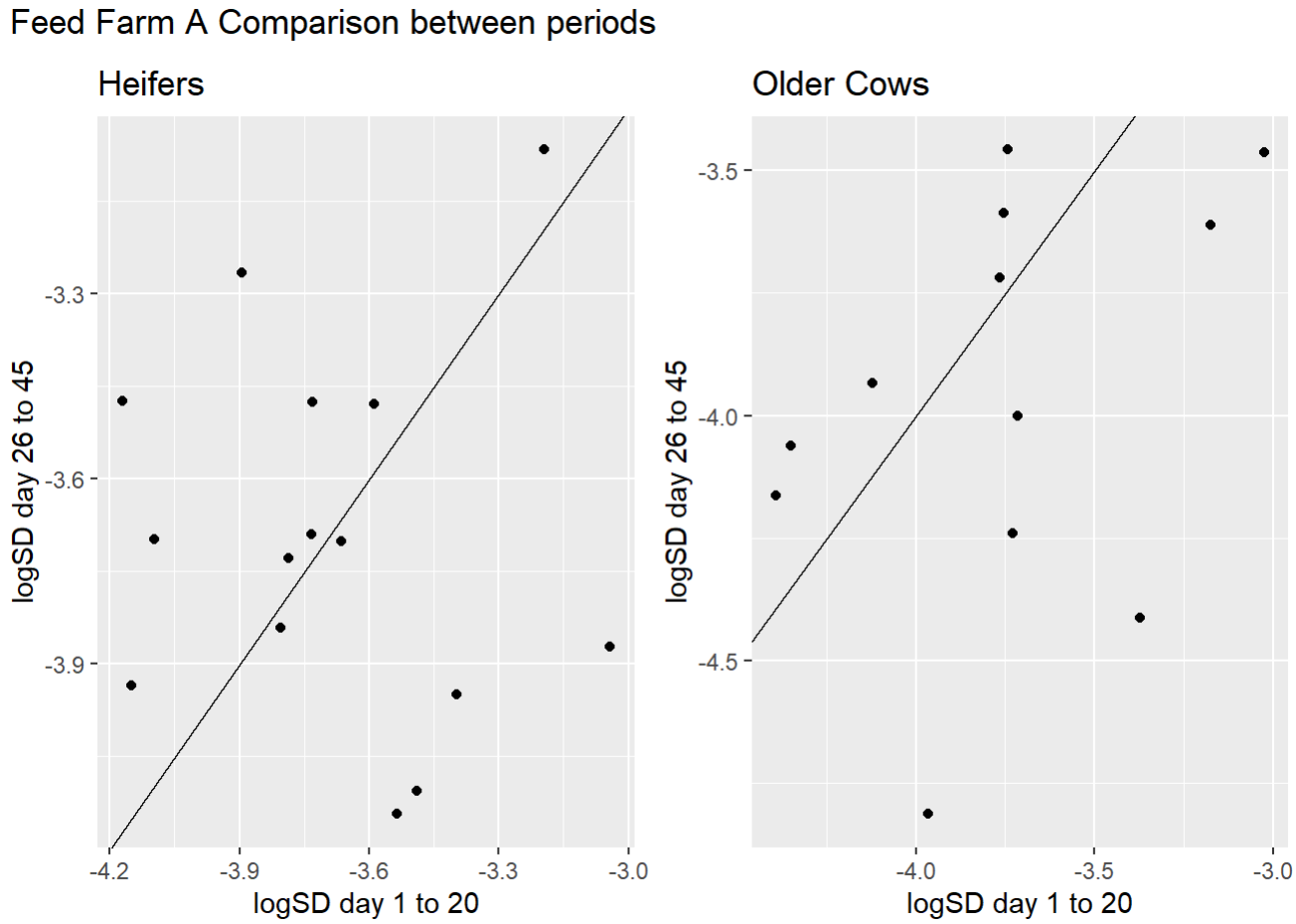
```
 Cubicle
Difference Period 1: P-value= 0.642
Difference Period 2: P-value= 0.82
Difference between Period 1 and 2 for Heifers: P-value= 0.974
```




```
 Alley
Difference Period 1: P-value= 0.986
Difference Period 2: P-value= 0.858
Difference between Period 1 and 2 for Heifers: P-value= 0.885
```



Feed
Difference Period 1: P-value= 0.623
Difference Period 2: P-value= 0.083
Difference between Period 1 and 2 for Heifers: P-value= 0.932



Farm B

The same analyses and plots for Farm B

```
farm="Farm B"
#hours.threshold = 16
CowDIM100 <- read.csv(paste0(data.path,"CowDIM100Wim.txt"), sep="")
CowDIM100 <- filter_summarise(CowDIM100)
```

```
`summarise()` has grouped output by 'Area'. You can override using the
`.groups` argument.

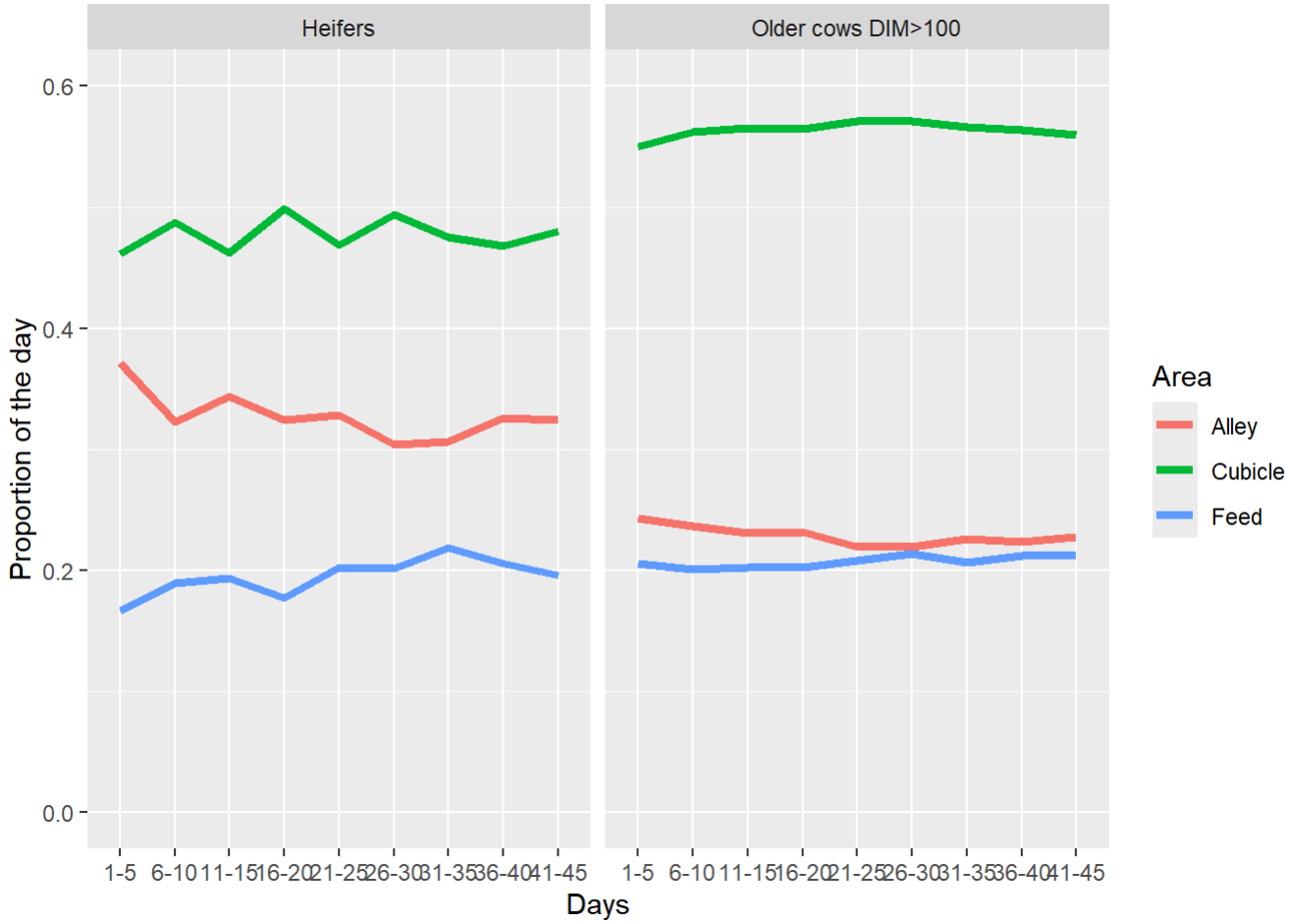
# A tibble: 6 × 4
# Groups:   Area [3]
  Area    Period avg_time_per_day proportion_NA
  <chr>    <dbl>            <dbl>         <dbl>
1 Alley        1            0.236             0
2 Alley        2            0.224             0
3 Cubicle      1            0.561             0
4 Cubicle      2            0.565             0
5 Feed         1            0.203             0
6 Feed         2            0.211             0
```

```
Heifers<- read.csv(paste0(data.path,"BudgetData_Dutch_w_hours.txt"), sep="")
Heifers <- filter_summarise(Heifers)
```

```
`summarise()` has grouped output by 'Area'. You can override using the
`.groups` argument.

# A tibble: 6 × 4
# Groups:   Area [3]
  Area    Period avg_time_per_day proportion_NA
  <chr>    <dbl>            <dbl>         <dbl>
1 Alley        1            0.341        0.05
2 Alley        2            0.315        0.0267
3 Cubicle      1            0.477        0.05
4 Cubicle      2            0.479        0.0267
5 Feed         1            0.182        0.05
6 Feed         2            0.205        0.0267
```

Make plots to get an overall picture per group (i.e. Heifers and older cows):



Overall GAMLSS modell

```
Results_GAMLSS =Overall_GAMLSS(Heifers, farm)
```

```
Warning in summary.gamlss(mod.group): summary: vcov has failed, option qr is used instead

[1] "########################################"
[1] "Cubicle"
[1] "Estimates of nuisance parameters not shown"
Mu link function:  identity
Mu Coefficients:
(Intercept)                          0.4863042  0.0636267   7.643 1.11e-13
```

```
n_intro                              -0.0032083  0.0035907  -0.893  0.37203
hours                                 0.0006400  0.0023970   0.267  0.78960
(Intercept)                          ***
n_intro
hours
Sigma link function:  log
Sigma Coefficients:
(Intercept) -2.043650   0.473725  -4.314 1.89e-05 ***
day          0.001767   0.002117   0.835    0.404
n_intro     -0.011611   0.038382  -0.303    0.762
hours       -0.030541   0.019923  -1.533    0.126

Warning in summary.gamlss(mod.group): summary: vcov has failed, option qr is used instead

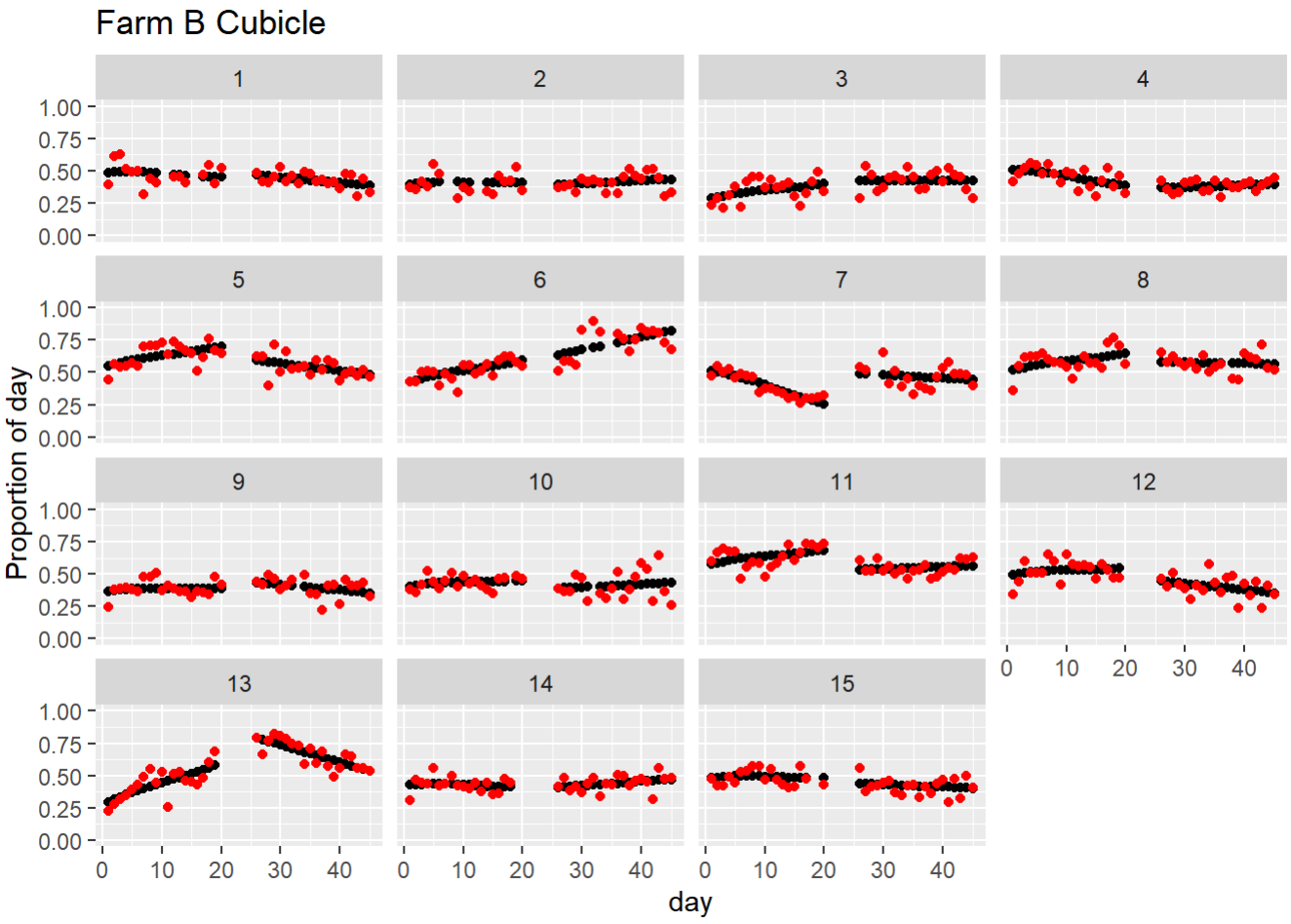
[1] "########################################"
[1] "Feed"
[1] "Estimates of nuisance parameters not shown"
Mu link function:  identity
Mu Coefficients:
(Intercept)                           2.412e-01  2.937e-02   8.211 2.35e-15
n_intro                              -3.589e-03  1.642e-03  -2.186 0.029308
hours                                -1.986e-03  1.113e-03  -1.784 0.075117
(Intercept)                          ***
n_intro                              *
hours                                .
Sigma link function:  log
Sigma Coefficients:
(Intercept) -2.612680   0.473725  -5.515 5.28e-08 ***
day          0.001328   0.002117   0.628   0.5306
n_intro      0.055451   0.038382   1.445   0.1491
hours       -0.042918   0.019923  -2.154   0.0316 *

Warning in summary.gamlss(mod.group): summary: vcov has failed, option qr is used instead

[1] "########################################"
[1] "Alley"
[1] "Estimates of nuisance parameters not shown"
Mu link function:  identity
Mu Coefficients:
(Intercept)                           0.2663958  0.0478850   5.563 4.41e-08
n_intro                               0.0060775  0.0031252   1.945 0.052404
hours                                 0.0019797  0.0016883   1.173 0.241536
(Intercept)                          ***
n_intro                              .
hours
Sigma link function:  log
Sigma Coefficients:
(Intercept) -3.047983   0.473725  -6.434 2.62e-10 ***
day         -0.001671   0.002117  -0.789    0.430
n_intro     -0.052333   0.038382  -1.363    0.173
hours        0.011622   0.019923   0.583    0.560
```

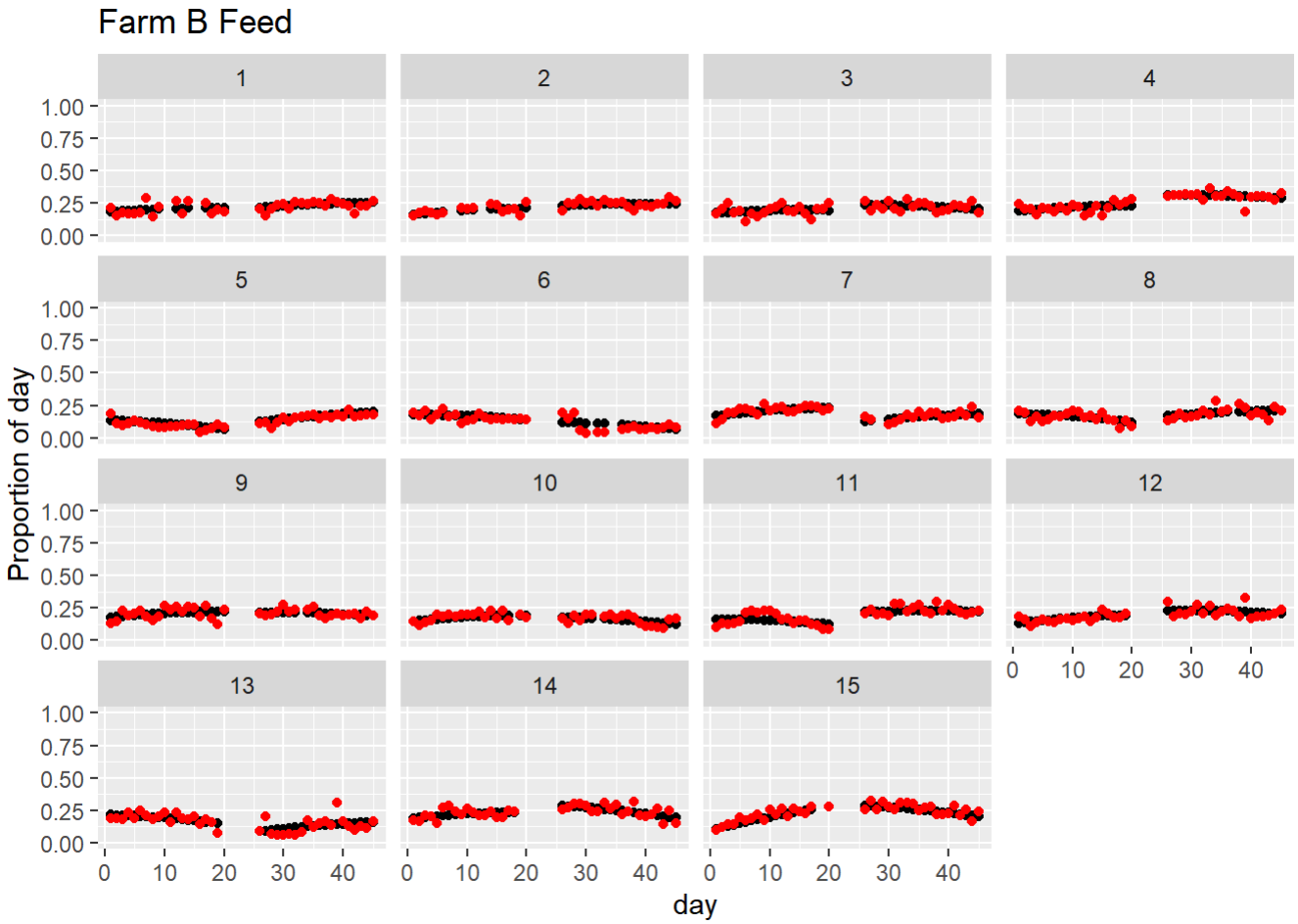
```
plot_GAMLSS_Results(Results_GAMLSS, Heifers, farm)
```

```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```

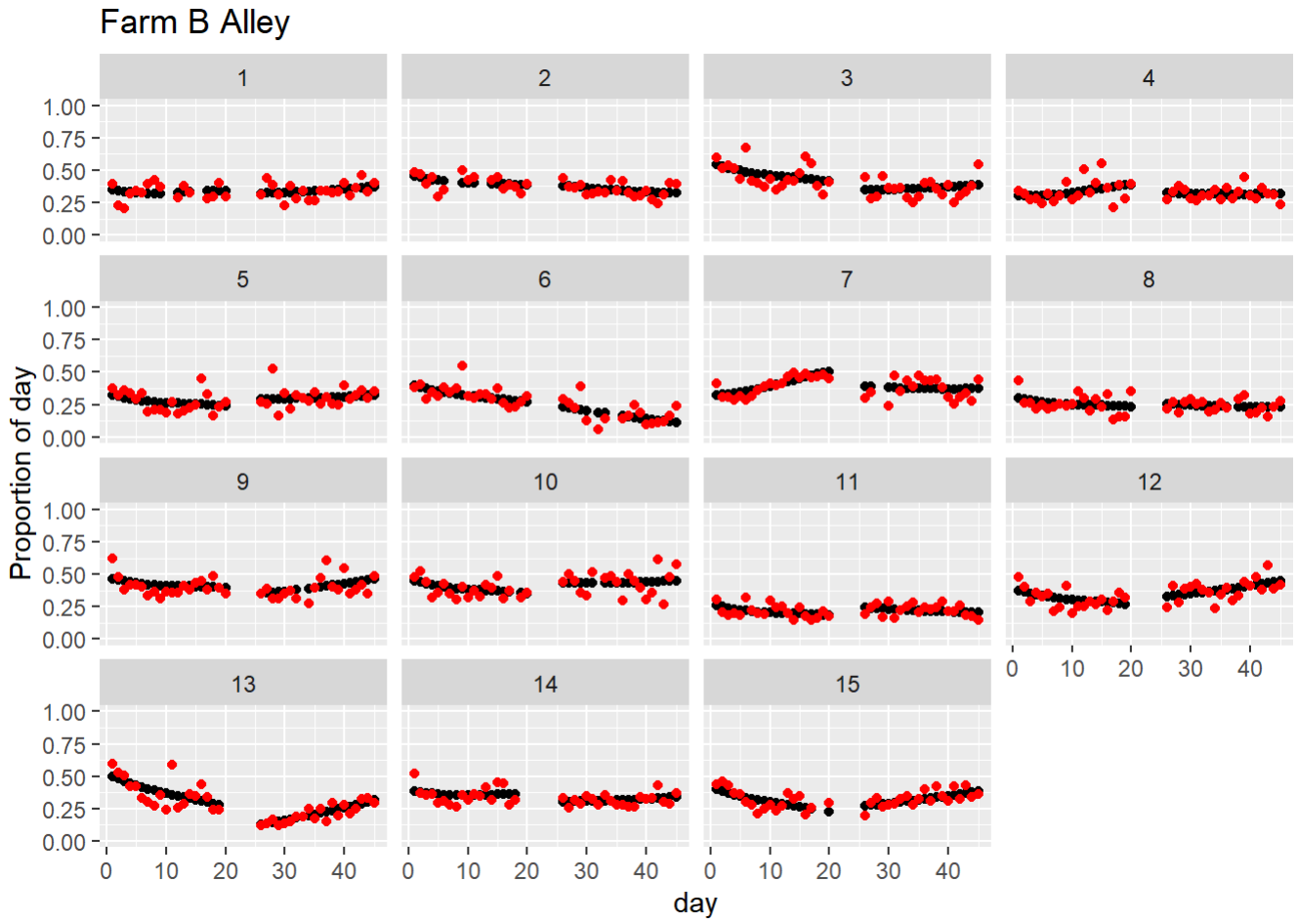



```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```



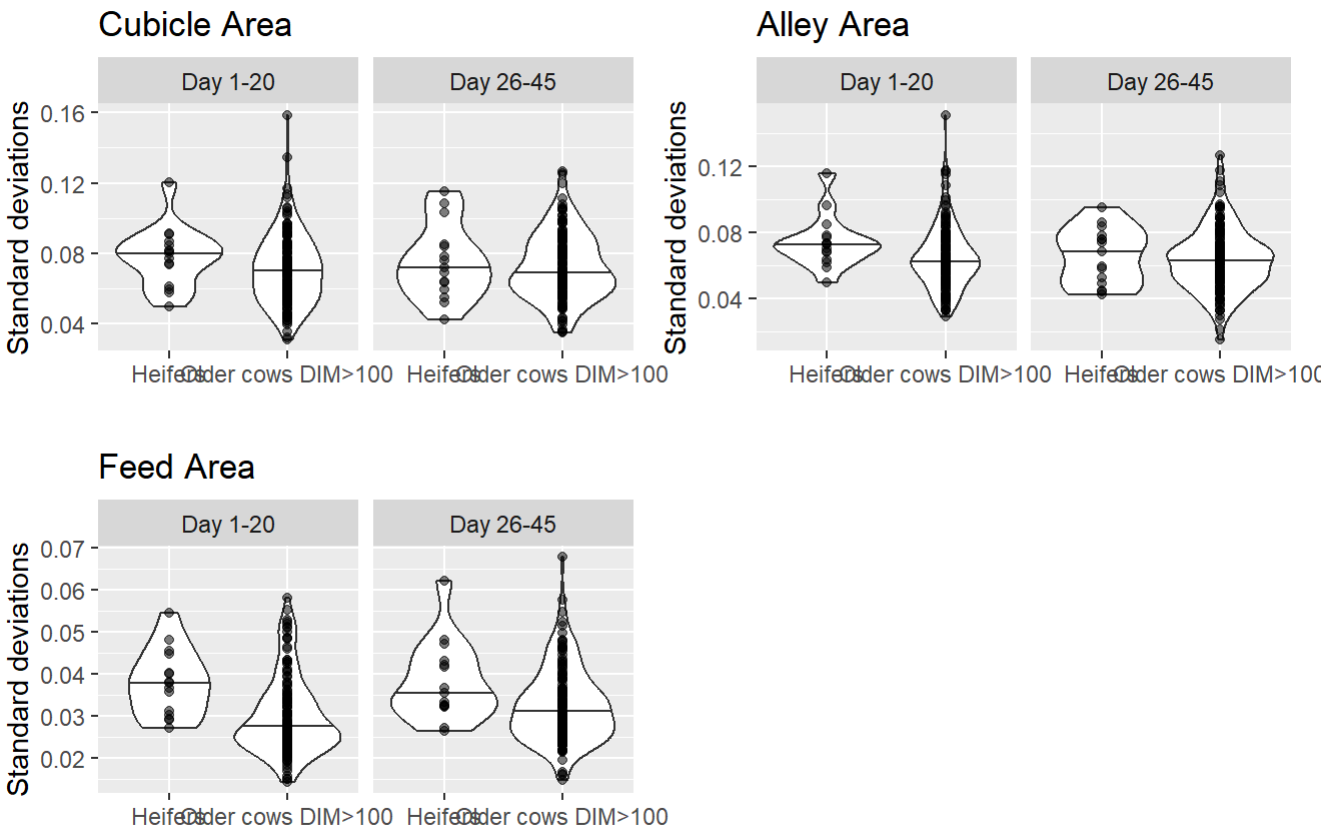

```
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
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New way of prediction in random()  (starting from GAMLSS version 5.0-6)
New way of prediction in random()  (starting from GAMLSS version 5.0-6)
```



Violin plots

Here we compute the variation of time spent in an area type between days (for a specified time period) for each individual separately and then plotting these as a violin plot:

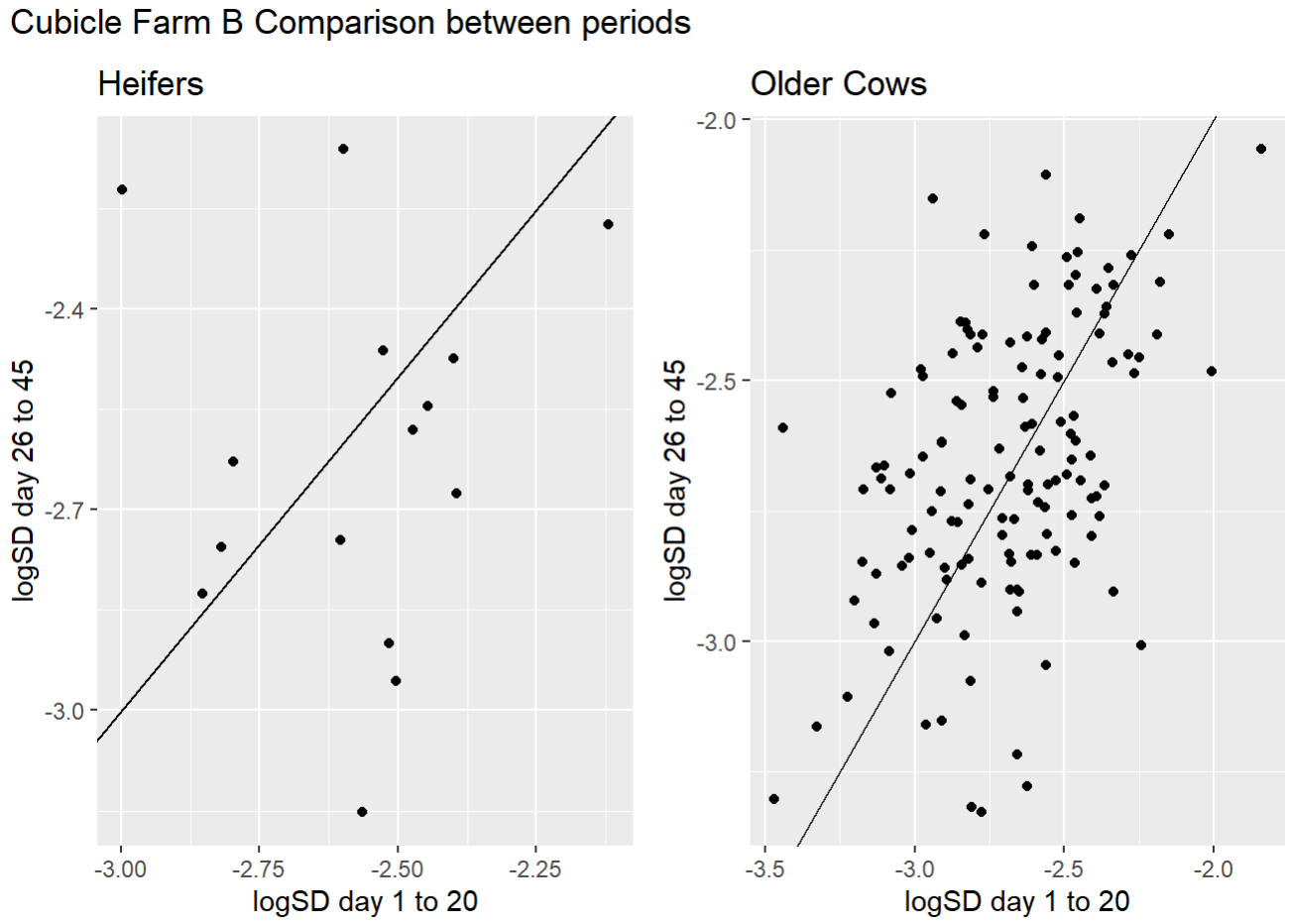
Farm B



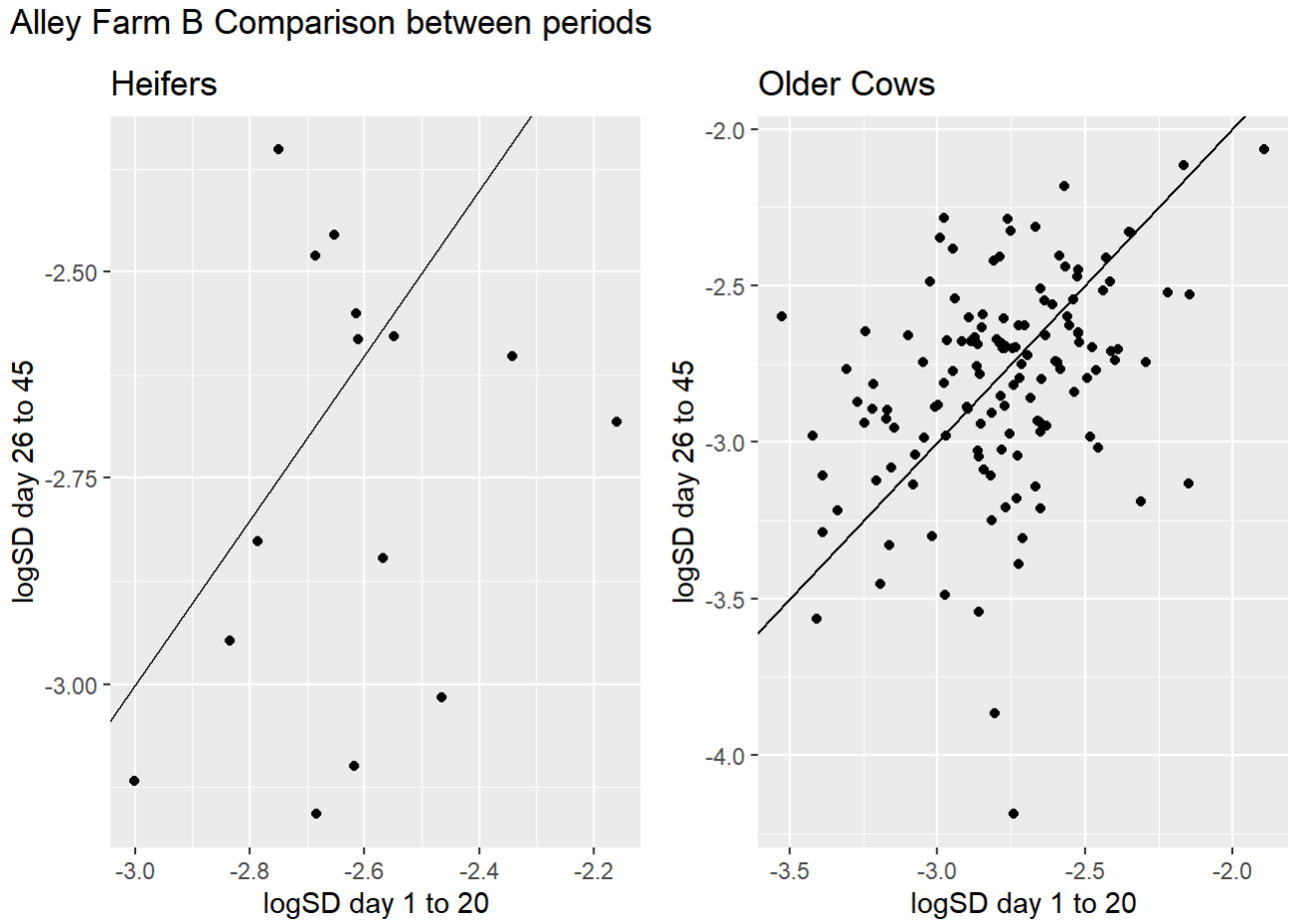
Analysis and plots of standard deviations

Here we analyze and plot the standard deviation of the raw data. We use log-transform to make it more normally distributed.

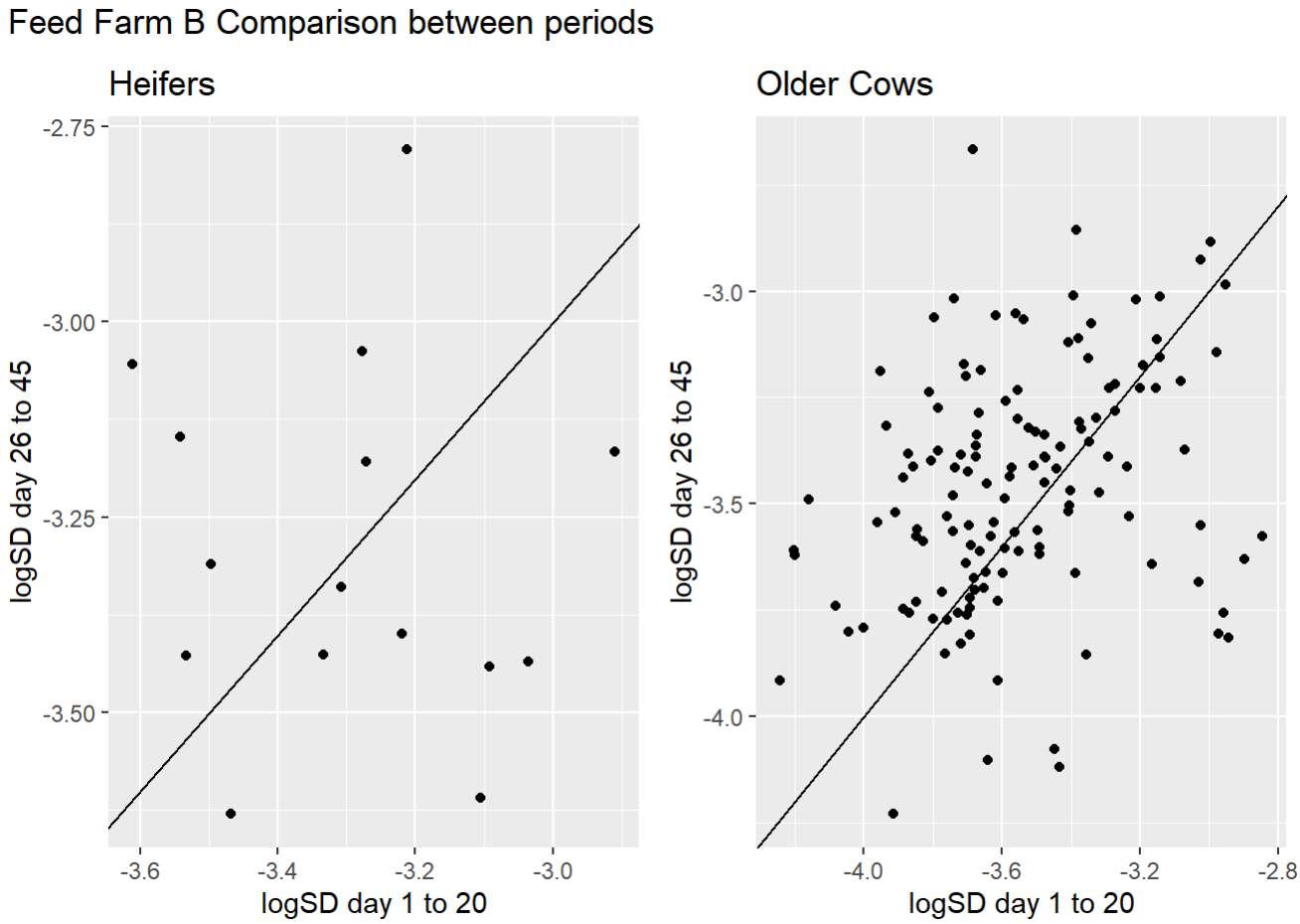
```
 Cubicle
Difference Period 1: P-value= 0.089
Difference Period 2: P-value= 0.663
Difference between Period 1 and 2 for Heifers: P-value= 0.584
```




```
 Alley
Difference Period 1: P-value= 0.011
Difference Period 2: P-value= 0.465
Difference between Period 1 and 2 for Heifers: P-value= 0.102
```

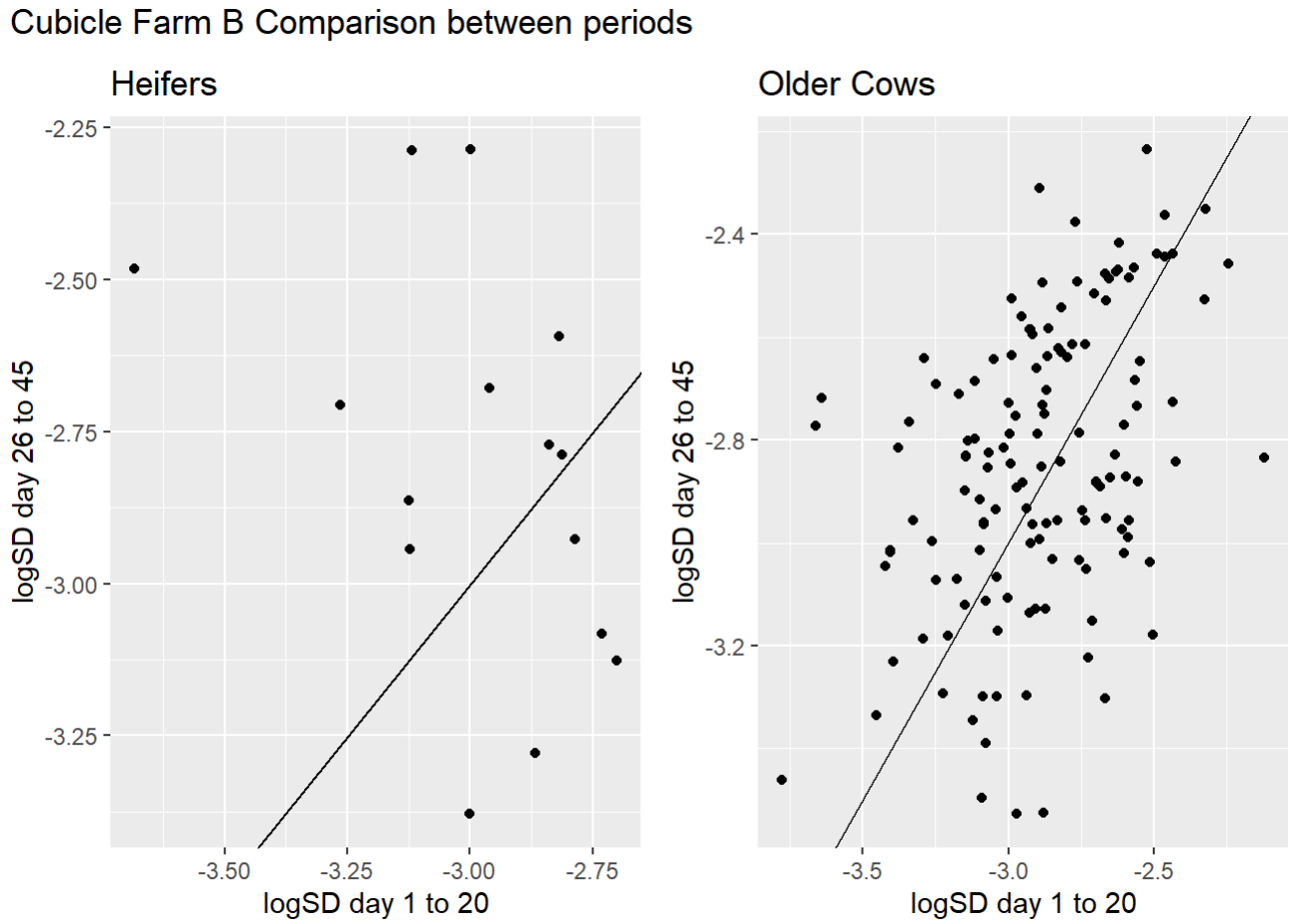



```
 Feed
Difference Period 1: P-value= 0.000300099
Difference Period 2: P-value= 0.017
Difference between Period 1 and 2 for Heifers: P-value= 0.981
```

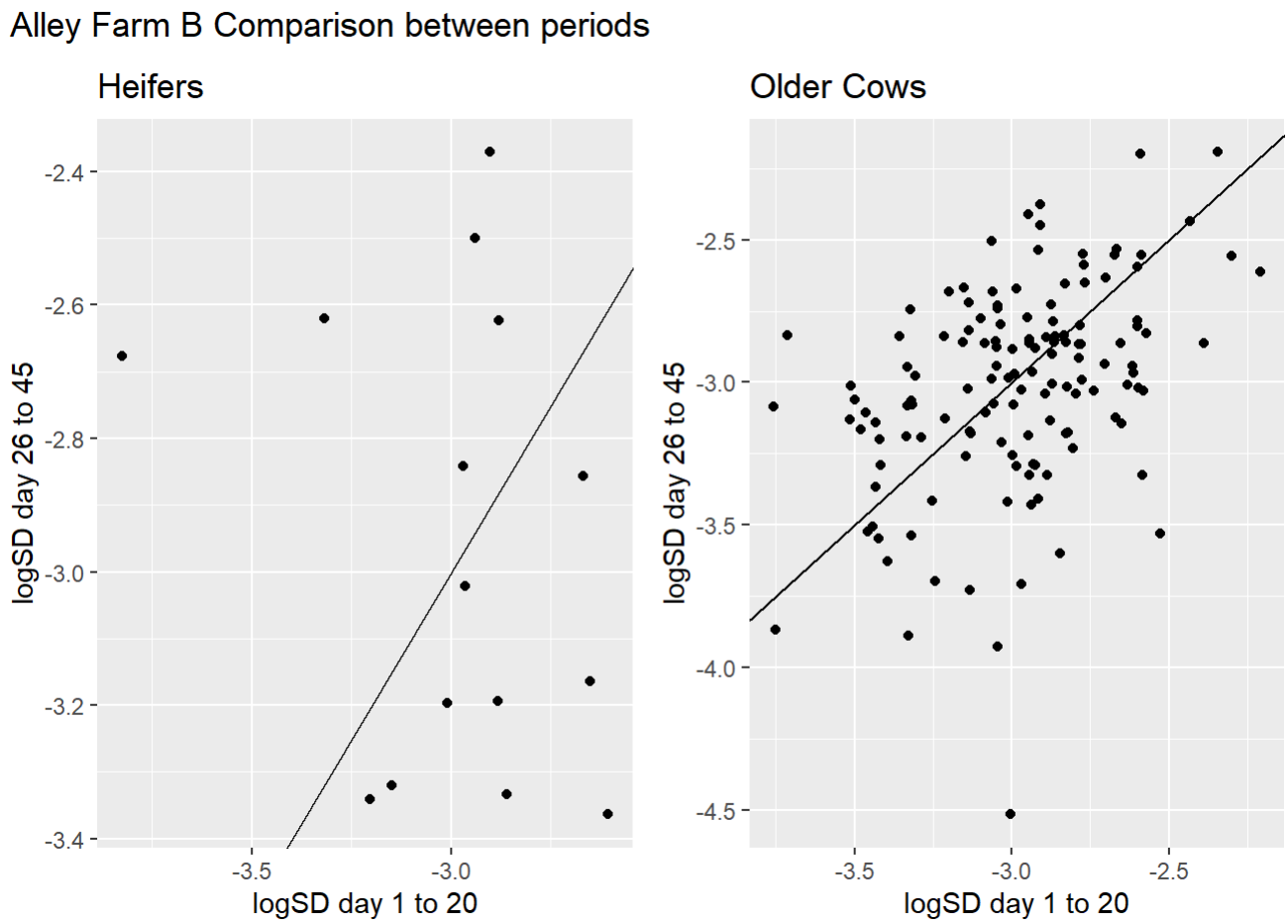


SD of the residuals:

```
 Cubicle
Difference Period 1: P-value= 0.182
Difference Period 2: P-value= 0.695
Difference between Period 1 and 2 for Heifers: P-value= 0.187
```




```
 Alley
Difference Period 1: P-value= 0.971
Difference Period 2: P-value= 0.604
Difference between Period 1 and 2 for Heifers: P-value= 0.843
```




```
 Feed
Difference Period 1: P-value= 0.172
Difference Period 2: P-value= 0.404
Difference between Period 1 and 2 for Heifers: P-value= 0.366
```

Feed Farm B Comparison between periods

